

Xinya Du

CONTACT	✉ Email: xinyadu8@gmail.com 🌐 Website Google Scholar GitHub	
RESEARCH INTERESTS	Large Language Models, LLM Alignment, LLM Reasoning, LLM Research Agent.	
EDUCATION	Cornell University Ph.D. in Computer Science (M.S. degree granted in Aug 2019) Worked with Claire Cardie (primary advisor) and Sasha Rush	Aug 2016 – Aug 2021
	Shanghai Jiao Tong University B.E. in Computer Science and Engineering Outstanding Graduates Award	Sep 2012 – Aug 2016
PROFESSIONAL EXPERIENCE	University of Texas at Dallas <i>Assistant Professor in Computer Science</i>	Aug 2022 – Present
	University of Illinois at Urbana-Champaign <i>Postdoctoral Research Associate. Worked with Heng Ji.</i>	Sep 2021 – Aug 2022
INDUSTRIAL EXPERIENCE	Google AI Research <i>Research Intern. Supervisor: Luheng He, Qi Li, Emily Pitler.</i>	May 2020 – Aug 2020
	Allen Institute for Artificial Intelligence (AI2) <i>Research Intern. Supervisor: Bhavana Dalvi, Peter Clark.</i>	Sep 2018 – Dec 2018
	Microsoft Research (MSR) <i>Research Intern. Supervisor: Paul Bennett, Ahmed Awadallah.</i>	May 2018 – Aug 2018
SELECTED AWARDS & HONORS	Best Paper Award	AAAI AI4Research, 2025
	NSF CAREER award	National Science Foundation, 2024
	Cisco Research Award	Cisco, 2024
	Amazon Research Award	Amazon, 2023
	Spotlight Rising Star in Data Science	University of Chicago, 2021
	Most Influential ACL Papers (15 each year)	Paperdigest, 2017
	National Scholarship (Top 1% students nationwide)	SJTU, 2013
	National Olympiad in Mathematics in Provinces, First Prize	2011
PUBLICATIONS	Note: † indicates that I am a co-leading author. * indicates equal contributions. You can also find my publication list on [DBLP] and [Google Scholar] pages.	

- [1] **HiPRAG: Hierarchical Process Rewards for Efficient Agentic Retrieval Augmented Generation** [\[pdf\]](#)
Peilin Wu, Mian Zhang, Kun Wan, Wentian Zhao, Kaiyu He, **Xinya Du**, Zhiyu Chen
In International Conference on Learning Representations (ICLR), 2026.
- [2] **Is Grokking Worthwhile? Functional Analysis and Transferability of Generalization Circuits in Transformers** [\[pdf\]](#)
Kaiyu He, Mian Zhang, Peilin Wu, **Xinya Du**, Zhiyu Chen
In Findings of the Association for Computational Linguistics: (ACL), 2026.
- [3] **Your Reasoning Model is Secretly a Reward Model – Optimization-Free Verification from Experience** [\[pdf\]](#)
Zhenwen Liang, Ruosen Li, Yujun Zhou, Linfeng Song, Dian Yu, **Xinya Du**, Haitao Mi, Dong Yu
In Annual Meeting of the Association for Computational Linguistics (ACL), 2026.
- [4] **FIFA: Unified Faithfulness Evaluation Framework for Text-to-Video and Video-to-Text Generation** [\[pdf\]](#)
Liqiang Jing, Viet Dac Lai, Seunghyun Yoon, Trung Bui, **Xinya Du**
In Findings of the Association for Computational Linguistics: (ACL), 2026.
- [5] **Dr.V: A Hierarchical Perception-Temporal-Cognition Framework to Diagnose Video Hallucination by Fine-Grained Spatial-Temporal Grounding** [\[pdf\]](#)
Meng Luo, Shengqiong Wu, Liqiang Jing, Tianjie Ju, Li Zheng, Jinxiang Lai, Tianlong Wu, **Xinya Du**, Jian Li, Siyuan Yan, Jiebo Luo, William Yang Wang, Hao Fei, Mong-Li Lee, Wynne Hsu
In International Journal of Computer Vision (IJCV), 2026.
- [6] **LMR-Bench: Evaluating LLM Agent’s Ability on Reproducing Language Modeling Research** [\[pdf\]](#)
Shuo Yan, Ziming Luo, Zimu Wang, Ruochen Li, Daoyang Li, Liqiang Jing, Kaiyu He, Peilin Wu, Juntong Ni, George Michalopoulos, Yue Zhang, Ziyang Zhang, Mian Zhang, Zhiyu Chen, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.
- [7] **FG-PRM: Fine-grained Hallucination Detection and Mitigation in Language Model Mathematical Reasoning** [\[pdf\]](#)
Ruosen Li, Ziming Luo, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2025.
- [8] **SciEvent: Benchmarking Multi-domain Scientific Event Extraction** [\[pdf\]](#)
Bofu Dong, Pritesh Shah, Sumedh Sonawane, Tiyaasha Banerjee, Erin Brady, **Xinya Du**, Ming Jiang
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.
- [9] **Search Wisely: Mitigating Sub-optimal Agentic Searches By Reducing Uncertainty** [\[pdf\]](#)
Peilin Wu, Mian Zhang, Xinlu Zhang, **Xinya Du**, Zhiyu Chen
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

- [10] **SFT or RL? An Early Investigation into Training R1-Like Reasoning Large Vision-Language Models** [\[pdf\]](#)
Hardy Chen, Haoqin Tu, Fali Wang, Hui Liu, Xianfeng Tang, **Xinya Du**, Yuyin Zhou, Cihang Xie
In Transactions on Machine Learning Research (TMLR), 2025.
- [11] **FGAIF: Aligning Large Vision-Language Models with Fine-grained AI Feedback** [\[pdf\]](#)
Liqiang Jing, **Xinya Du**
In Transactions on Machine Learning Research (TMLR), 2025.
- [12] **LDC: Learning to Generate Research Idea with Dynamic Control** [\[pdf\]](#)
Ruochen Li, Liqiang Jing, Chi Han, Jiawei Zhou, **Xinya Du**
In AAAI 2025 AI4Research Workshop. (Oral)
Best Paper Award at AAAI AI4Research.
- [13] **VistaDPO: Video Hierarchical Spatial-Temporal Direct Preference Optimization for Large Video Models** [\[pdf\]](#)
Haojian Huang*, Haodong Chen*, Shengqiong Wu, Meng Luo, Jinlan Fu, **Xinya Du**, Hanwang Zhang, Hao Fei
In International Conference on Machine Learning (ICML), 2025.
- [14] **DSBench: How Far Are Data Science Agents to Become Data Science Experts?** [\[pdf\]](#)
Liqiang Jing, Zhehui Huang, Xiaoyang Wang, Wenlin Yao, Wenhao Yu, Kaixin Ma, Hongming Zhang, **Xinya Du**, Dong Yu
In International Conference on Learning Representations (ICLR), 2025.
- [15] **FIHA: Autonomous Hallucination Evaluation in Vision-Language Models with Davidson Scene Graphs** [\[pdf\]](#)
Bowen Yan, Zhengsong Zhang, Liqiang Jing, Eftekhari Hossain and **Xinya Du**
In Findings of the Association for Computational Linguistics: (ACL), 2025.
- [16] **Efficient Document-level Event Relation Extraction**
Ruochen Li, Zimu Wang, **Xinya Du**
In 10th Workshop on Representation Learning for NLP at Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2025.
- [17] **PRD: Peer Rank and Discussion Improve Large Language Model-based Evaluations** [\[pdf\]](#)
Ruosen Li, Teerth Patel, **Xinya Du**
In Transactions on Machine Learning Research (TMLR), 2024.
- [18] **Large Language Models for Automated Open-domain Scientific Hypotheses Discovery** [\[pdf\]](#)
Zonglin Yang, **Xinya Du**[†], Junxian Li, Jie Zheng, Soujanya Poria, Erik Cambria
In Findings of the Association for Computational Linguistics: (ACL), 2024.
Best poster award in ICML AI4Science, 2024 (Top 1% of 250 submissions).
- [19] **Multi-hop Event-centric Question Answering with Explanations**
Ruosen Li, Zimu Wang, Son Quoc Tran, Lei Xia, **Xinya Du**
In Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

- [20] **FaithScore: Fine-grained Evaluations of Hallucinations in Large Vision-Language Models** [\[pdf\]](#)
 Liqiang Jing, Ruosen Li, Yunmo Chen, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2024.
- [21] **IQA-EVAL: Automatic Evaluation of Human-Model Interactive Question Answering** [\[pdf\]](#)
 Ruosen Li, Ruochen Li, Barry Wang, **Xinya Du**
In Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.
- [22] **QAEvent: Event Extraction as Question-Answer Pairs Generation** [\[pdf\]](#)
 Milind Choudhary, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EACL), 2024.
- [23] **Language Models as Inductive Reasoners** [\[pdf\]](#)
 Zonglin Yang, Li Dong, **Xinya Du**[†], Hao Cheng, Erik Cambria, Xiaodong Liu, Jianfeng Gao, Furu Wei
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2024. (Oral)
- [24] **Making Natural Language Reasoning Explainable and Faithful** [\[pdf\]](#)
Xinya Du
In AAAI Conference on Artificial Intelligence (AAAI), 2024. (Oral)
- [25] **Proto-CLIP: Vision-Language Prototypical Network for Few-Shot Learning** [\[pdf\]](#)
 Jishnu Jaykumar P, Kamalesh Palanisamy, Yu-Wei Chao, **Xinya Du**, Yu Xiang
In International Conference on Intelligent Robots and Systems (IROS), 2024. (Oral)
- [26] **Document-level Causal Relation Extraction with Knowledge-guided Binary Question Answering** [\[pdf\]](#)
 Zimu Wang, Lei Xia, Wei Wang, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2024.
- [27] **Leveraging Structured Information for Explainable Multi-hop Question Answering and Reasoning** [\[pdf\]](#)
 Ruosen Li, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2023.
- [28] **Process of Elimination for Multiple Choice Reasoning** [\[pdf\]](#)
 Chenkai Ma, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.
- [29] **Probing Representations for Document-level Event Extraction** [\[pdf\]](#)
 Barry Wang, **Xinya Du**, Claire Cardie
In Findings of the Association for Computational Linguistics: (EMNLP), 2023.
- [30] **End-to-end Case-Based Reasoning for Commonsense Knowledge Base Completion** [\[pdf\]](#)
 Zonglin Yang, **Xinya Du**[†], Erik Cambria, Claire Cardie
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2023. (Oral)

- [31] **Zero-Shot Classification by Logical Reasoning on Natural Language Explanations** [\[pdf\]](#)
Chi Han, Hengzhi Pei, **Xinya Du**, Heng Ji
In Findings of the Association for Computational Linguistics: (ACL), 2023.
- [32] **Toward Consistent and Informative Event-Event Temporal Relation Extraction** [\[pdf\]](#)
Xiaomeng Jin, Haoyang Wen, **Xinya Du**, Heng Ji *In MATCHING at Annual Meeting of the Association for Computational Linguistics (ACL), 2023. (Oral)*
- [33] **Logical Entity Representation in Knowledge-Graphs for Differentiable Rule Learning** [\[pdf\]](#)
Chi Han, Qizheng He, Charles Yu, **Xinya Du**, Hanghang Tong, Heng Ji
In International Conference on Learning Representations (ICLR), 2023.
- [34] **RESIN-11: Schema-guided Event Prediction for 11 Newsworthy Scenarios** [\[pdf\]](#)
Xinya Du, Zixuan Zhang, Sha Li, Heng Ji, the RESIN team
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): System Demonstrations, 2022.
Top ranking system in DARPA KAIROS evaluation. [\[Link\]](#)
- [35] **Retrieval-Augmented Generative Question Answering for Event Argument Extraction** [\[pdf\]](#)
Xinya Du, Heng Ji
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2022.
- [36] **Dynamic Global Memory for Document-level Argument Extraction** [\[pdf\]](#)
Xinya Du, Sha Li, Heng Ji
In Annual Meeting of the Association for Computational Linguistics (ACL), 2022.
- [37] **Automatic Error Analysis for Document-level Information Extraction** [\[pdf\]](#)
Aliva Das*, **Xinya Du***, Barry Wang*, Kejian Shi, Jiayuan Gu, Thomas Porter, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2022.
- [38] **Template Filling with Generative Transformers** [\[pdf\]](#)
Xinya Du, Alexander M. Rush, Claire Cardie
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021.
- [39] **GRIT: Generative Role-filler Transformers for Document-level Event Entity Extraction** [\[pdf\]](#) **Xinya Du**, Alexander M. Rush, Claire Cardie
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2021. (Oral)
- [40] **Few-shot Intent Classification and Slot Filling with Retrieved Examples** [\[pdf\]](#)
Dian Yu, Luheng He, Yuan Zhang, **Xinya Du**, Panupong Pasupat, Qi Li *In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021. (Oral)*

- [41] **QA-Driven Zero-shot Slot Filling with Weak Supervision Pretraining** [\[pdf\]](#)
Xinya Du, Luheng He, Qi Li, Dian Yu, Panupong Pasupat, Yuan Zhang
In Annual Meeting of the Association for Computational Linguistics (ACL), 2021.
- [42] **Event Extraction by Answering (Almost) Natural Questions** [\[pdf\]](#)
Xinya Du, Claire Cardie
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020.
(Oral)
Top 1% most cited articles published in Computer Science in 2020.
- [43] **Improving Event Duration Prediction via Time-aware Pre-training** [\[pdf\]](#)
Zonglin Yang, **Xinya Du**, Alexander M. Rush, Claire Cardie
In Findings of the Association for Computational Linguistics: (EMNLP), 2020.
- [44] **Document-Level Event Role Filler Extraction using Multi-Granularity Contextualized Encoding** [\[pdf\]](#)
Xinya Du, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2020.
- [45] **Leveraging Structured Metadata for Improving Question Answering on the Web** [\[pdf\]](#)
Xinya Du, Adam Fourney, Robert Sim, Claire Cardie, Paul Bennett, Ahmed Hassan Awadallah
In Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL/IJCNLP), 2020.
- [46] **Be Consistent! Improving Procedural Text Comprehension using Label Consistency** [\[pdf\]](#)
Xinya Du, Bhavana Dalvi, Niket Tandon, Antoine Bosselut, Wen-tau Yih, Peter Clark, Claire Cardie
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2019.
- [47] **Harvesting Paragraph-Level Question-Answer Pairs from Wikipedia** [\[pdf\]](#)
Xinya Du, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2018. **(Oral)**
Top 1% most cited articles published in Computer Science in 2018.
- [48] **Identifying Where to Focus in Reading Comprehension for Neural Question Generation** [\[pdf\]](#)
Xinya Du, Claire Cardie
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017.
- [49] **Learning to Ask: Neural Question Generation for Reading Comprehension** [\[pdf\]](#)
Xinya Du, Junru Shao, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2017.
Most Influential ACL Papers (15 each year). [\[Link\]](#)
Featured in New Scientist and TechRepublic. [\[Link\]](#)
Top 0.1% most cited articles published in Computer Science in 2017.

- [50] **Cornell Belief and Sentiment System at TAC 2016** [\[pdf\]](#)
Vlad Niculae, Kai Sun, Xilun Chen, Yao Cheng, **Xinya Du**, Esin Durmus, Arzoo Katiyar,
Claire Cardie
In Text Analysis Conference (TAC), 2016.

PROFESSIONAL **Grant-Proposal Review Panelists**

SERVICES NSF Panel in 2026 (for IIS/RI Future CORe program).

NSF Panel in 2025 (for IIS/RI MEDIUM program).

Chairing Roles:

NeurIPS Area Chair, 2026.

ACL Demo Track Area Chair, 2026.

ICLR Areal Chair, 2025.

ACL (Annual Meeting of the Association for Computational Linguistics) Rolling Review
Area Chair, 2024.

ACL Area Chair, 2023.

ACL Session Chair, 2024.

EMNLP (Conference on Empirical Methods in Natural Language Processing) Area Chair,
2024. (**Outstanding Area Chair Award**).

EMNLP Demo Track Area Chair, 2024.

NAACL (Conference of the North American Chapter of the Association for Computational
Linguistics) Website Chair, 2024.

COLING (International Conference on Computational Linguistics) Senior Area Chair,
2024, 2025.

Seminar/Conference/Workshop/Tutorial Organizing:

Frontiers of LLMs for Automated Scientific Discovery: A Tutorial

Organizing team: Zonglin Yang, Chandan Reddy, Xinya Du and Mengdi Wang.

In AAAI Conference on Artificial Intelligence (AAAI), 2026

AI4Research: Towards a Unified Knowledge-grounded Scientific Research Lifecycle.

Organizing team: Qingyun Wang, Wenpeng Yin, Lifu Huang, Yi R. Fung, Xinya Du, Carl
Edwards, Tom Hope.

In AAAI Conference on Artificial Intelligence (AAAI), 2025.

Evaluations and Benchmarks in Context of Multimodal LLM.

Organizing team: Hao Fei, Xiang Yue, Kaipeng Zhang, Long Chen, Jian Li, Xinya Du.

*In The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),
2025.*

The First Workshop on Multimodal Knowledge and Language Modeling

Organizing team: Liqiang Jing, Xinya Du, Hao Fei, Jing Gu, Manling Li, Aixin Sun,
William Wang

In International Joint Conference on Artificial Intelligence (IJCAI), 2025.

Journal Reviewer:

TKDE (IEEE Transactions on Knowledge and Data Engineering).

TASLP (IEEE Transactions on Audio, Speech and Language Processing).

TNNLS (IEEE Transactions on Neural Networks and Learning Systems).

TALLIP (ACM Transactions on Asian and Low-Resource Language Information Processing).

TKDD (ACM Transactions on Knowledge Discovery from Data).

CL (Computational Linguistics).

KAIS (Knowledge and Information Systems).

AI Communication.

IPM (Information Processing and Management).

Conference Committee Member:

ACL (Annual Meeting of the Association for Computational Linguistics).

NeurIPS (Conference on Neural Information Processing Systems).

ICLR (International Conference on Learning Representations).

AISTATS (International Conference on Artificial Intelligence and Statistics).

EMNLP (Conference on Empirical Methods in Natural Language Processing).

NAACL (Conference of the North American Chapter of the Association for Computational Linguistics).

AACL/IJCNLP (Asia-Pacific Chapter of the Association for Computational Linguistics / International Joint Conference on Natural Language Processing).

IJCAI (International Joint Conference on Artificial Intelligence).

AAAI (AAAI Conference on Artificial Intelligence).

NLPCC (Natural Language Processing and Chinese Computing).

CoNLL (Conference on Computational Natural Language Learning).

WNUT (Workshop on Noisy User-generated Text).

MRQA (Workshop on Machine Reading for Question Answering).

SEM (Joint Conference on Lexical and Computational Semantics).

PhD/MS Committee Member:

PhD: Kangshuo Li, Souradeep Nanda, Ali Riahi, Xiangci Li, Jishnu Jaykumar Padalunkal, Basel Abdeen, Yibo Hu.

MS: Wooseong Yang, Shubham Patel.

Other Activities:

Faculty advisor and judge: Association for Computing Machinery (ACM) Symposium, UT Dallas, 2023–present.

Faculty Advisor: Women Who Compute (WWC), UT Dallas, 2024–present.

Faculty Advisor: Society of Asian Scientists and Engineers (SASE), UT Dallas, 2024 – present.

Invited Speaker: UTD AI Society’s “AI Chats”, 2025.

Panelist: FuturED 2024 Workshop on Future of Event Detection, 2024.

Faculty Host: K-12 Outreach, UT Dallas, Summer 2023, 2024.

Research Mentor: Research, Inquiry, Design Experience (RIDE) Project, UT Dallas, 2024.

PhD Admission Committee Member, UT Dallas CS Department, 2022, 2023, 2024.

Site Host: North American Computational Linguistics Olympiad (NACLO), 2023, 2024.

PhD Admission Committee Member, Cornell CS Department, 2021.

Volunteer: Cornell CS Department PhD Visit Day, 2019, 2020, 2021.

Student Volunteer: ACL, 2017, 2018.

Student Volunteer: EMNLP, 2017.

RECENT TALKS

Pushing the Boundaries of LLM Agents in Data Science and Machine Learning Research
Nvidia, Dec 2025.

Inside LLMs: Knowledge, Alignment, and Agents.

Samsung Research America (SRA), May 2025. .

Towards Autonomous Scientific Discovery with Large Language Models.

UT Southwestern AI in Medicine Seminar Series, May 2025.

Synergizing Knowledge and Large Language Models.

University of Illinois Urbana-Champaign, Data Mining Group Seminar, Oct 2024.

University of Massachusetts–Amherst, Machine Learning & Friends Lunch, Sep 2024.

Synergy between Large Language Model and Knowledge.

University of North Texas, Oct 2024.

Panel at Future of Event Detection Workshop.

Conference on Empirical Methods in Natural Language Processing (EMNLP), Nov 2024.

Faculty Round Table Talk.

UT Dallas Hobson Wildenthal Honors College, Aug 2024.

Large Language Models: Knowledge, Reasoning and Factuality.

Samsung Electronics America, Mar 2024.

Making Natural Language Reasoning Explainable and Faithful.

AAAI Conference on Artificial Intelligence (AAAI), Jan 2024.

Open-ended Evaluations of Foundational Models: Alignment and Faithfulness.

Shanghai Jiao Tong University, Computer Science Department Seminar, Dec 2023.

Fudan University, Computer Science Department Seminar, Dec 2023.

ChatGPT: Fact vs. Fiction.

UT Dallas, Forum sponsored by The Dallas Morning News, May 2023.

Towards More Intelligent Extraction of Information from Documents.

Allen Institute for Artificial Intelligence, Mar 2022.

Facebook, Mar 2022.

Simon Fraser University, Feb 2022.

Rensselaer Polytechnic Institute, Feb 2022.

University of California, Merced, Feb 2022.

Hong Kong University of Science and Technology, February 2022.

Towards More Intelligent Extraction of Information from Documents.

UIUC, Siebel School of Computing and Data Science Speaker Series, Feb 2022.

Towards More Informed Extraction of Events from Documents.

University of Chicago, Rising Stars in Data Science Workshop, Jan 2021.

Tencent AI Research America, Nov 2020.

Event Extraction by Answering (Almost) Natural Question.

UIUC, Information Extraction and Knowledge Acquisition Class, Sep 2020.

LwLL: Progress on the NLP Front.

Cornell University, DARPA site visit, Apr 2020.